

Fixed cost remains the same, fixed cost p.u changes
Variable cost changes, variable cost p.u remains same

2.

Flexible Budget

Particulars	50%	60%	70%	90%
<u>Fixed Expenses</u>				
Salaries	50,000	50,000	50,000	50,000
Rent & Taxes	40,000	40,000	40,000	40,000
Depreciation	60,000	60,000	60,000	60,000
Admin Exp	70,000	70,000	70,000	70,000
<u>Variable Exp</u>				
Materials	2,00,000	2,40,000	2,80,000	3,60,000
Labour	2,50,000	3,00,000	3,50,000	4,50,000
Others	40,000	48,000	56,000	72,000
<u>Semi-Variable exp</u>				
Repairs	1,00,000	1,00,000	1,10,000	1,15,500
Indirect Labour	1,50,000	1,50,000	1,65,000	1,73,250
Others	90,000	90,000	99,000	1,03,950
TOTAL COST	10,50,000	11,48,000	12,80,000	14,94,700
SALES	-	11,00,000	13,00,000	15,00,000
Profit (Loss)		(48,000)	30,000	5300

3.

Flexible Budget

Particulars	50%	60%	70%
<u>Variable Overheads</u>			
Indirect Labour	8750	10,500	12,250
Indirect Materials	7000	8400	9800
<u>Semi-Variable o/h</u>			
Repairs & Maintenance	6650	7000	7350
Electricity	23,100	25,200	27,300
<u>Fixed overheads</u>			
Office expenses	70,000	70,000	70,000
Insurance	4000	4000	4000
Depreciation	20,000	20,000	20,000
Estimated direct labour	1,00,000	1,20,000	1,20,000
TOTAL COST	2,03,950		
TOTAL COST	1,39,500	1,45,100	1,50,700
Labour hours	1,00,000	1,20,000	1,40,000
Overhead recovery rate/hr	1.395	1.209	1.076

4.

Flexible budget

Particulars	60%		80%		100%	
	Total	P.U	Total	P.U	Total	P.U
Materials	60,000	100	80,000	100	1,00,000	100
Labour	24,000	40	32,000	40	40,000	40
Direct Labour	6000	10	8000	10	10,000	10
Factory o/h						
Fixed Admin expenses	16,000	26.67	16,000	20	16,000	16
Variable	24,000	40	32,000	40	40,000	40
Admin expense						
Fixed	18,000	30	18,000	22.5	18,000	18
Variable	12,000	20	16,000	20	20,000	20
Total Cost	1,60,000	267	2,02,000	253	2,44,000	244

Cash Budgets

5. Working Notes:

Sales	Feb	Mar	Apr	May	June
Feb 70K	17,500	52,500			
Mar 80K		20,000	60,000	22,000	
Apr 96K			24,000	67,500	
May 1L				25,000	75,000
Jun 1.2L					30,000
Total	17,500	72,500	84,000	97,000	1,05,000

Cash budget for the months from
April to June 2008

Particulars	Apr (Rs)	May (Rs)	June (Rs)
Opening Cash Balance	32,000	57,000	82,000
Revenue:			
Cash Sales (25%)	24,000	25,000	30,000
Credit Sales received from debtors (75%)	60,000	72,000	75,000
Total (A)	1,16,000	1,54,000	1,87,000
Payments:			
Payment to Suppliers (2m credit)	(44,000)	(56,000)	(60,000)
Wages (1m credit)	(9,000)	(9,000)	(11,000)
Expenses (1m)	(6,000)	(7,000)	(9,000)
Income tax	-	-	(28,000)
Total (B)	(59,000)	(72,000)	(1,08,000)
Closing Balance (A-B)	57,000	82,000	79,000

Cash

6. Working Notes: Calculation of collection from Debtors

Sales	Feb	March	Apr	May	Jun
Feb 1.80L	60,000	960,000	960,000		
Mar 1.92L			960,000	96,000	
Apr 1.08L				540,000	540,000
May 1.74L					87,000
Jun 1.26L					

Cash Budget

Particulars	April	May	June
Opening balance	25,000	56,000	(47,000)
Receipts:			
Sales [Received from debtors]	1,86,000	1,50,000	1,41,000
Total Receipts (A)	2,11,000	2,06,000	94,000
Payments			
Payment to suppliers	(1,44,000)	(2,43,000)	(2,46,000)
Wages paid	(11,000)	(10,000)	(15,000)
Total Payments (B)	(1,55,000)	(2,53,000)	(2,61,000)
Closing Balance (A-B)	56,000	(47,000)	(1,67,000)

Negative balance indicates overdraft facility required by the company.

7. Working Note 1: Calculation of collection from Debtors

Month	Feb	Mar	Apr	May	June
Feb	96K	48,000	48,000		
Mar	1,04K		52,000	52,000	
Apr	64K			32,000	32,000
May	92,800				46,400
Jun	70,400				

Working Note 2: Calculation of wages

Month	Feb	Mar	Apr	May	Jun
Feb	2500	2500			
Mar	3000		3000		
Apr	2000			2000	
May	2500				2500
Jun	2000				

Cash Budget

Particulars	April	May	June
Opening Balance	5000	5680	(7084)
Receipts:			
Cash Sales [Sales Amt $\times \frac{201}{200}$]	15680	22,736	17,248
Cash received from debtors	1,00,000	84,000	78,400
Income from investments	5000	-	-
Total (A)	1,25,680	1,12,146	88,564
Payments			
Payment to creditors	(1,00,000)	(1,04,000)	(1,06,000)
Miscellaneous exp	(5000)	(6000)	(12,000)
Wages paid in cash	7500(6000)	9000(7500)	(6000)
O/S wages paid	(3000)	(2000)	(2500)
Rent paid	(3000)	-	-

Income Tax			(25,000)
Total Expenses (B)	(1,20,000)	(1,19,500)	(1,26,500)
Closing balance (A-B)	5680	(7084)	(37926)

Production = Sales + C/S - O/P
(Budgeted) (Budgeted) (Desired) - O/P

10. Production Budget for year ended
(Budgeted Production) 30/06/2010

Particulars	A	B	C
Budgeted Sales	1,50,000	1,00,000	70,000
Add: Desired C/S	15,000	14,500	8000
less: Opening Stock	(14,000)	(5000)	(8000)
Budgeted Production	1,51,000	1,09,500	70,000

8. Production Budget for half-year
ended 30/09/2002 [in units]

Particulars	Apr Mar	May	Jun	Jul	Aug	Sep
Gamma						
Budgeted Sales	900	1100	1400	1800	2200	2200
Add: Desired C/S	550	700	900	1100	1100	900
Total	1450	1800	2300	2900	3300	3100
less: O/S	450	550	700	900	1100	1100
Budgeted Production	1000	1250	1600	2000	2200	2000

∴ Total units produced by Gamma = 10,050

	Delta	Apr	May	Jun	July	Aug	Sep
Budgeted Sales		2900	2900	2500	2100	1700	1700
Add: Desired c/s		1450	1450	1050	850	850	950
less: O/S		1450	1450	1250	1050	850	850
Desired Production		2900	2700	2300	1900	1700	1800
		= 13,300					

Summarised Production Cost budget for half year ended 30/09/2002

Particulars	Gamma (10050 units)		Delta (13,300 units)	
	CU	Total Cost	CU	Total Cost
Material	50	5,02,500	80	10,64,000
Labour	20	2,01,000	30	3,99,000
Other Expenses	10	1,00,500	15	1,99,500
Total	80	8,04,000	125	16,62,500

9. Quantitative cum Financial Budget for sales of the current year

Zones	Snow Cream (SP12)		Talcum Powder (SP14)		Gold Cream (SP15)	
	No of units	Sales	No of units	Sales	No of units	Sales
Zone A	4,00,000	48,00,000	3,00,000	42,00,000	3,25,000	52,00,000
Zone B	2,75,000	33,00,000	4,20,000	58,80,000	3,00,000	48,00,000
Total	6,75,000	81,00,000	7,20,000	1,00,80,000	6,25,000	1,00,00,000

11. Calculation of Budgeted Production:

$$\text{BUDGETED PRODUCTION} = \text{BUDGETED SALES} + \text{C/S} - \text{O/S}$$

$$= 40,000 + 7000 - 5000$$

$$= 42,000 \text{ units}$$

Calculation of materials consumed:

$$\text{Material A} = 42,000 \times 3 = 1,26,000$$

$$\text{Material B} = 42,000 \times 4 = 1,68,000$$

$$\text{Materials procured} = \text{Materials Consumed} + \text{C/L} - \text{O/P}$$

Material procurement Budget (Quantitative)

Particulars	Mat A	Mat B	Mat A	Material B
Materials Cons			1,26,000	1,68,000
Add: C/L of Mat	15,000			
Material in hand	15,000	25,000		
Materials on order	8000	10,000	23,000	35,000
less: O/P of Mat				
Mat in hand	(12,000)	(20,000)		
Mat on order	(7000)	(11,000)	(19,000)	(31,000)
to be Mat _x procured			1,30,000	1,72,000

$$12. \text{Budgeted production} = 50,000 + 4000 - 10,000$$

$$= 54,000 \text{ units}$$

Calculation of materials consumed:

$$\text{Mat A} = 54,000 \times 2 = 1,08,000$$

$$\text{Mat B} = 54,000 \times 3 = 1,62,000$$

Material Procurement Budget [Quantitative]

Particulars	Material A	Material B
Budgeted Cons	1,08,000	1,62,000
Add: Closing Stock	13,000	16,000
Less: Opening Stock	(12,000)	A (15,000)
Budgeted production	1,09,000	1,63,000

13. Working Notes:

Volume	1,20,000	1,50,000
Indirect Mat	2,64,000	3,30,000
Cost p.u	2.2	2.2 → Variable

Indirect labour	1,50,000	1,87,500
Cost p.u	1.25	1.25 → Variable

Maintenance	84,000	1,02,000
Cost p.u	0.7	0.68 → Semi-Variable
	$1,02,000 - 84,000$	0.6
	$1,50,000 - 1,20,000$	

$$\therefore \text{Fixed} = [0.1 \times 1,20,000] = 12,000$$

Supervision	1,98,000	2,34,000
Cost p.u	1.65	1.56
	$2,34,000 - 1,98,000$	1.2
	$1,50,000 - 1,20,000$	

$$\text{Variable proportion of semi variable cost} = \frac{\Delta \text{ cost}}{\Delta \text{ volume of production}}$$

Factory o/h budget (production 1,40,000)

Particulars	Amt:
Indirect Materials (@ 2.2/unit)	308,000
Indirect labour (@ 1.25/unit)	1,75,000
Maintenance:	
Fixed:	
Variable: $[0.6 \times 1,40,000]$	12,000
Supervision	84,000
Fixed:	
Variable: $[1.2 \times 1,40,000]$	54,000
Engineering Service	1,68,000
Total cost	94,000
	8,95,000

MASTER BUDGET

Particulars	(₹)	(₹)
Sales	A 40,00,000	
Less: Cost of Goods Sold	B 60,00,000	1,00,00,000
Direct Materials Cost (40% of Sales)	40,00,000	
Direct Wages $[15 \times 12,000 \times 12]$	21,60,000	
Factory o/h:		
Works manager & Foreman (20,000 x 12)	2,40,000	
Foreman Salary $[15,000 \times 12]$	1,80,000	
Stores & Spares $[1,00,00,000 \times 2.5\%]$	2,50,000	
Depreciation	1,26,000	
Light & power	50,000	
Repairs & Maintenance	80,000	
Misc. Expenses	2,16,000	
Admin, selling & Distribution Exp	1,40,000	(74,42,000)
Profit		25,58,000

15. Calculation of cost/unit [Total]

Raw material cost/unit		C	
D	45	1	45
E	15	10	150
F	15	2	30
G	5	8	40
Total		10	50
For A		265(A)	335(B)
		295(C)	

Calculation of Labour cost/unit

Labour		Product A		Product B		Product C	
Labour rate/hr	Product A	Product B	Product C	Labour rate/hr	Product A	Product B	Product C
Skilled	6	8	36	4	24	3	18
Unskilled	5	8	40	6	30	6	30
Total		76	54	48			

Variable of h cost plv 9 11 7

Total Cost plv 350 400 350

Calculation of no of units sold

	A	B	C
Sales Ratio (Given)	x	2x	4x
Selling Price p.u	450	550	650
Variable Cost	(350)	(400)	(350)
Contribution	100	150	300
Fixed Cost			
Total Contribution	100x	300x	1200x
Fixed Cost	(15,35,000)	(5,30,000)	(8,45,000)
EBIT (Total)	1,20,00,000/12		
16002 - (30,00,000) = 1,20,00,000/12			
2875 units x = 2500			

i) Sales Budget

Particulars	A	B	C
Sales Ratio	2	2x	4x
Sales Qty	2500	5000	10,000
Selling price p.u	450	550	650
Total Budgeted Sales	11,25,000	27,50,000	65,00,000

PRODUCTION BUDGET

Particulars	A	B	C
Budgeted Sales	2500	5000	10,000
Add: Closing Stock	450	900	2000
less: Opening Stock	(500)	(1000)	(3000)
Budgeted Production	2450	4900	9700

PARTS USAGE [PROCUREMENT] BUDGET

Products	Net units	D	E	F	G
		Net units	Total	Net units	Total
A	2450	1	2450	10	24500
B	4900	1	4900	2	9800
C	9700	1	9700	6	58200
			17,050	92,500	1,70,500

PROCUREMENT BUDGET

Particulars	D	E	F	G
Mat. Consumed	17,050	92,500	1,70,500	88,000
Add: Closing Stock	1350	900	18,000	9000
less: Opening Stock	(1500)	(1000)	(20,000)	(10,000)
Mat. Procured	16900	92,400	1,68,500	87,000
Purch price p.u	45	15	15	5
Purchase Value	2,60,500	15,86,000	25,27,500	4,35,000

e) Manpower Budget

Products - Eign	Production in units	skilled		Unskilled	
		No of hrs	Total hrs	No of hrs	Total hrs
A	2450	6	14,700	8	19,600
B	4900	4	19,600	6	29,400
C	9700	3	29,100	6	58,200
			63,400		1,07,200
			6		5
Wage per hr :			380400		5,36,000
Total wages					

f) Master Budget (monthly)

Particulars	(₹)	(₹)	(₹)
Sales (11,25,000 + 27,50,000 + 65,00,000)			
Product A B C			1,03,75,000
(-) Variable Costs			
Materials Consumed			
D (17,1050 X 45)		7,67,250	
E (92,500 X 15)		13,87,500	
F (1,70,500 X 15)		25,57,500	
G (88,000 X 5)		4,40,000	(51,52,250)
(-) Labour			
Skilled		3,80,400	
Unskilled		5,36,000	(9,16,400)
(-) Overheads			
A (2450 X 9)		22,050	
B (4900 X 11)		53,900	
C (9700 X 7)		67,900	(1,43,850)
Goods sold from Opening Stock			
(Sales - Production)			
A - (500 X 350)		17,500	
B - (1000 X 400)		400,000	

C - (300 X 350)		1,05,000	(162,500)
Contribution			40,00,000
(-) Fixed Cost			
A - 15,75,000			(30,00,000)
B - 5,80,000			
C - 8,45,000			
Profit (monthly)			10,00,000
X	X	X	X

4. RELEVANT COSTING

→ Decision Making

Total Relevant Cost

Materials

Already available

N/A

Regularly used?

Out of pocket cost
It is a relevant cost

Yes

No

Opportunity Cost

Replacement cost

→ Scrap

→ NRV

→ Substitute

Relevant Cost

2.

Calculation of Relevant Cost of materials

Materials	Nature & Computation	Amts (£)
A	1000 units to be purchased (Here out of pocket cost is the relevant cost) @ £6 p.u	6000
B	1000 units to be purchased (600 to be replaced by 400 units to be purchased) @ £5 p.u	5000
C	700 units @ NRV, here it is 700×2.5	1750
	300 units @ Replacement cost 300×4	1200
D	Substitute for Mat E 300 units $\times 5 = 1500$	
	200 units @ NRV i.e. 200 units of £7.50 = 1500	1500
TOTAL RELEVANT COST		15,950

We can accept the order given that other costs don't exceed £6550.

[The relevant cost of material is less than the income from the order]

3.

Computation of Relevant Cost of Machine

Nature & Computation

(£)

Variable cost of operating the machine is relevant 60,000

Reduction in NRV as opportunity cost is relevant 15,000

TOTAL RELEVANT COST 75,000

4.

Computation of Relevant Cost of Special order

Name

Nature & Computation

(£)

Equipment

Reduction in NRV as opportunity cost is relevant (1600-800)

700

Rent

Rent for storage in warehouse (out of pocket)

18,000

Sub-assembly

The cost of making is $(10+6+6) = 22$ & cost of buying is £24. Better when in-house

22 p.u

5.

Computation of Opportunity Cost

Particulars	X	Y	Z
Selling price	10	12	12
V.C.	6	9	7
Contribution	4	3	5
Effective production (Less of demand & production)	2000	2000	400
Contribution (Total)	8000	6000	4500
Opportunity Cost	{6000}	{8000}	{8000}

6. Particulars	Make	Buy & Sell	Buy & Hire out
Variable Cost	1,80,000	1,90,000	1,90,000
Relevant Benefit	-	10,000 x 2 20,000	24,000
	1,80,000	1,70,000	1,69,000

Buy & Hiring out is the less option as it gives the lowest Net Relevant Cost.

7. Particulars	Nature & Computation	Am't
Material M	Material M is a non moving item $\therefore$ NRV	80

Labour: Permanent employment is committed cost. Hence irrelevant.

Out of pocket expense: It is a relevant cost 30.

Minimum price to be charged 110 ₹

8. Particulars	Nature & Computation
Relevant cost	Operating Cost of old mach ₹100/hr
	Cost of new machine ₹35,00,000
	Operating cost of new mach. ₹.70/hr
	NRV of old machine ₹12,00,000
	Revenue from both mach ₹120/hr
	Difference b/w OC = 50

No of hrs of the new machine used: Relevant benefit = $35L - 12L = 23L$

Contribution/hr = $\frac{23L}{50 \text{ hrs}} = 46,000$

Break-even point: $BEP \text{ of Mach 1} = \frac{30L}{20} = 1,00,000 \text{ hrs}$

BEP for mach 2 = $\frac{35L}{50} = 70,000 \text{ hrs}$

MODULE 2: Marginal Costing & Absorption Costing

cost volume

CVP Analysis: Examines changes in cost [fixed & variable] & sales affects the business profitability

It is primarily used to calculate the Break-even point & to forecast the exact no. of units a co. needs to sell to achieve a desired profit

Key concepts & components

1. Marginal Cost Equation:

Sales	XXX
(-) V.C	(XX)
Contribution	XXX
(-) F.C	(XX)
Profit/loss	XXX

$$\begin{aligned} \text{a) Contribution} &= \text{Sales} - \text{Variable Cost} \\ &= \text{F.C} + \text{Profit} \end{aligned}$$

2. Profit Volume Analysis: The proportion of sales available to cover fixed cost & profit

$$\text{PV Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

$$= \frac{\text{Contribution/unit}}{\text{Sales/unit}} \times 100$$

$$= \frac{\Delta \text{Contribution}}{\Delta \text{Sales}} \times 100$$

$$\text{assuming FC} = 0 = \frac{\Delta \text{Profit}}{\Delta \text{Sales}} \times 100$$

2. Variable Cost ratio is given

$$\text{VCR} = \frac{\text{VC}}{\text{Sales}} \times 100$$

3. Break-Even Point: Level of sales where, Contribution is equal to fixed Cost. Hence, here, there is neither profit nor loss.

$$\text{BEP (in Sales Value)} = \frac{\text{FC} \times \text{Sales}}{\text{Contribution}} \quad \text{or} \quad \frac{\text{FC}}{\text{PV Ratio}}$$

4. Margin of Safety: It represents the difference b/w actual sales & BEP sales

$$\text{MOS} = \text{Actual Sales} - \text{BEP Sales}$$

$$\text{MOS} = \frac{\text{Profit}}{\text{Contribution}} \times \text{Sales} = \frac{\text{Profit}}{\text{PV Ratio}}$$

$$\text{MOS (units)} = \frac{\text{Profit}}{\text{Contribution/unit}}$$

$$\text{MOS as \% of Sales} = \frac{\text{AS} - \text{BEPs} \times 100}{\text{Sales}}$$

$$\Rightarrow \frac{\text{MOS} \times 100}{\text{Sales}}$$

2. FC = 5,00,000
VC = 10/unit
SP = 15/unit
output = 1,50,000

Profit:

	Value	1 unit
Sales =	22,50,000	15

V.C	(15,00,000)	10
-----	-------------	----

Contribution = 7,50,000

(-) FC (5,00,000)
2,50,000

3. Contribution = Sales - V.C
= 80,000 - 60,000

BEP = 60,000

BEP = FC

Contribution

60,000 = FC

FC = 15,000

Profit = Contribution - FC
= 20,000 - 15,000
= 5,000

4. Contribution = 18 - 12 = 6

FC = 60,000

BEP = $\frac{\text{FC}}{\text{Cont/unit}} = \frac{60,000}{6} = 10,000 \text{ units}$

Value = 10,000 x 18 = 1,80,000

5. ~~BEP = 9000~~

~~Contribution = 25~~

~~FC = 2,70,000~~

~~BEP = $\frac{2,70,000}{25} = 10,800 \text{ units} = 10,800 \times 18$~~

~~25~~

~~10,800~~

~~9000~~

~~Contribution 20~~

5. ~~BEP = 9000~~ = $\frac{2,70,000}{30}$

~~x = 75~~

~~x = 30 + 75~~

~~x = 105~~

7.

Particulars

Sales

Profit

Y₁

20L

8L

Y₂

30L

4L

PV Ratio = $\frac{\text{change in profit}}{\text{change in Sales}} \times 100$

= $\frac{2,00,000}{10,00,000} \times 100 = 20\%$

8. $PV \text{ Ratio} = \frac{\Delta \text{ Profit}}{\Delta \text{ Sales}} \times 100$
 $= \frac{4000}{10,000} \times 100$
 $= 40\%$

$PV R = \frac{\text{Contribution (x)}}{\text{Sales}}$

$\frac{40}{100} = \frac{21}{80,000}$

$\frac{40}{100} = \frac{x_2}{90,000}$

$x_1 = 32,000$

$x_2 = 36,000$

	Y_1	Y_2
Contribution	32,000	36,000
(-) F.C	22,000	22,000
Profit	10,000	10,000

$BEP = \frac{FC}{PV \text{ Ratio}} = \frac{22,000}{0.4} = 55,000$

9. Selling Price 10
 MC per unit 8
 Contribution / unit 2

$MOS = \frac{\text{Profit}}{\text{Contribution/unit}} \times SP$

$= \frac{20,000}{2} \times 10 = 1,00,000$

10.

Sales	1,00,000
(-) VC	60,000
(80K-20K)	
Contribution	40,000
(-) FC	20,000
Profit	20,000

a) $PV \text{ Ratio} = \frac{\text{Contribution}}{\text{Sales}} = \frac{40,000}{1,00,000} \times 100 = 40\%$

b) $BEP = \frac{FC}{PV \text{ Ratio}} = \frac{20,000}{40\%} = 50,000$

c) $Mos = \text{Actual sales} - BEP \text{ sales}$
 $= 1,00,000 - 50,000$
 $= 50,000$

11.

Particulars	a)	i)	ii)	iii)	iv)	v)	vi)
18,000 Sales	20,000	20,000	20,000	20,000	20,000	22,000	18,000
(9000) (-) VC	(10,000)	(10,000)	(10,000)	(9000)	(11,000)	(10,000)	(10,000)
9000 Contribution	10,000	10,000	10,000	11,000	9000	12,000	8000
(6000) (-) FC	(6000)	(4800)	(6600)	(6000)	(6000)	(7200)	(6000)
3000 Profit	4000	5200	3400	5000	3000	4800	2000
9000 PV ratio	10,000	10,000	10,000	11,000	9000	12,000	8000
18,000	20,000	20,000	20,000	20,000	20,000	22,000	18,000
50% 50%	50%	50%	50%	55%	45%	54%	44%
6000 BEP Sales	6000	4800	6600	6000	6000	7200	6000
0.5 0.5	0.5	0.5	0.5	0.55	0.45	0.54	0.44
12000	12,000	9600	13,200	10,909	13,333	13,333	13,626
18,000 MOS	20,000	20,000	20,000	20,000	20,000	22,000	18,000
(-) (2,000)	12,000	9600	13,200	10,909	13,333	13,333	13,626
6000	8000	10,400	6800	9091	6667	8667	4364

9
4-2-23
37
4/3

C-FC

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With ΔFC , PV ratio remains unaffected
FC - BEP - directly related to fixed cost
Fixed Cost - MoS - inversely related.

$$6. \therefore \text{Contribution} = \text{Sales} - \text{V.C} \\ = 10 - 5 \\ = 55$$

$$\therefore \text{BEP} = \frac{\text{FC}}{\text{Contribution/unit}} = \frac{2,00,000}{5} = 40,000 \text{ units}$$

If S.P is $\downarrow 10\%$, SP = 9

$$\therefore \text{Contribution} = 9 - 5 \\ = 45$$

$$\therefore \text{BEP} = \frac{2,00,000}{45} = 50,000 \text{ units}$$

$$12. a) \text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} = \frac{\text{Sales} - \text{VC}}{\text{Sales}} \\ = \frac{10,00,000 - 6,00,000}{10,00,000} = \frac{4,00,000}{10,00,000} = 0.4 \\ = 40\%$$

$$\text{Here FC} = \text{Contribution} - \text{Profit} \\ = 4,00,000 - 1,00,000 \\ = ₹ 3,00,000$$

$$b) \text{BEP} = \frac{\text{FC} \times \text{Sales}}{\text{Contribution}} = \frac{3,00,000 \times 25,000}{4,00,000} \\ = ₹ 18,75,000$$

desired Cu
for
sales

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c) Profit when sales = 12,00,000

$$\therefore \text{Increase in sales} = \frac{2,00,000}{10,00,000} \times 100\% = 20\%$$

$$\therefore \text{Increase in VC} = 6,00,000 + 20\% = 7,20,000$$

$$\therefore \text{Sales} = 12,00,000$$

$$(-) \text{V.C} \quad (7,20,000)$$

$$\text{Contribution} \quad 4,80,000$$

$$(-) \text{F.C} \quad (3,00,000)$$

$$\text{Profit} \quad \underline{1,80,000}$$

d) Sales when profit = 2,00,000

$$\text{Sales} \quad (12,50,000)$$

$$(-) \text{V.C} \quad (7,50,000)$$

$$\text{Contribution} \quad 5,00,000$$

$$(-) \text{F.C} \quad (3,00,000)$$

$$\text{Profit} \quad \underline{2,00,000}$$

$$\text{V.C as \% of sales} = \frac{6,00,000}{10,00,000} \times 100 = 60\%$$

$$\therefore \text{If sales} = x$$

$$\text{then Contribution} = x - 60\% \text{ of } x$$

$$5,00,000 = x - 0.6x$$

$$x = \frac{5,00,000}{0.4} = ₹ 12,50,000$$

$$= 12,50,000$$

$$\text{Sales to earn desired profit} = \frac{\text{FC} + \text{desired profit}}{\text{PV Ratio}}$$

13. i) P/V Ratio = $\frac{\text{Contribution}}{\text{Sales}} = \frac{50,000}{2,00,000} = 0.25 = 25\%$

ii) BEP = $\frac{FC}{\text{Contribution}} \times \text{Sales Value} = \frac{15,000}{50,000} \times 60,000 = 18,000$

iii) ~~$\frac{40}{100} \times 60,000 = 24,000$~~
 ~~$24,000 < 15,000$~~
 ~~$2,00,000 \times 0.25 = 50,000$~~
 ~~$2,00,000 \times 0.40 = 80,000$~~

~~Sales = 3,20,000~~

iv) ~~MOS @ 3,00,000 sales~~
 ~~$2,00,000 - 35,000$~~
 ~~$3,00,000 - ?$~~

~~50,500~~

iii) Sales = $\frac{50,000}{40\%} = 1,25,000$

iv) MOS = Sales - BEP
 $= 3,00,000 - 60,000$
 $= 2,40,000$

v) $\frac{3,00,000}{(2,25,000)}$
 $75,000$
 c) $\frac{15,000}{60,000}$

vi)

$\frac{2,20,000}{(1,65,000)}$
 $55,000$
 $(15,000)$
 $70,000$

3,40,000

$x + 0.75x = 55,000$
 $0.25x = 55,000$

vii)

$\frac{7,60,000}{(5,70,000)}$
 Contr 1,40,000
 FC (15,000)
 PBT 1,75,000
 Tax (1,05,000)
 $70,000$

let PBT be x.
 $x - 0.6x = 70,000$
 $0.4x = 70,000$
 $x = 1,75,000$

$1,40,000 = x - 0.75x$
 $1,40,000 = 0.25x$

14. At 100% Capacity,

Particulars	Co. 1	Co. 2	Merged
Capacity utilisation	100%	100%	100%
Sales	600	500	1100
Variable cost	(440)	(375)	815
Contribution	160	125	285
Fixed Cost	(80)	(50)	(130)
Profit	80	75	155

P/V ratio of merged Co. = $\frac{\text{Cont} \times 100}{\text{Sales}}$
 $= \frac{285}{1100} \times 100 = 25.91$

130 → 211
136.5 → 24.09

215

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Break Even Sales revenue of merged Co = $\frac{FC}{P/V \text{ Ratio}}$

= $\frac{130}{0.2591}$

= 501.736

Break Even Capacity of merged Co = $\frac{BE \text{ Sales}}{\text{Sales}} = \frac{501.74}{1100}$

= 45.61%

i) Profitability of the merged plant @ 80% Capacity.

Sales @ 80% Capacity	= 880
(-) V.C (80%)	= (652)
Contribution	228
(-) F.C	(130)
Profit	98

Profitability = $\frac{98}{880} \times 100 = 11.14\%$

iii) Sales required for 275L

791
1586
205
(130)
75

Target Sales = $\frac{FC + \text{Desired Profit}}{P/V \text{ Ratio}}$

iv) $\frac{65}{791} \times 100$ $\left[\frac{130 \times 5}{100} \right] 6.5$

Absorption Costing

Standard rule if $CS > OS$ [Inventory increases] - absorption costing profit is higher than Marginal costing profit

If $CS < OS$ [Inventory decreases] - absorption costing profit is lower than the marginal costing profit

No change in inventory, the profits under both methods are same.

11. a)

Particulars	Mill	Stream
Sales	2,00,000	1,50,000
(-) Variable Cost	(1,60,000)	(1,00,000)
Contribution	40,000	50,000

b)

Particulars	Mill	Stream
Sales	3,00,000	1,80,000
(-) Variable Cost	(2,40,000)	(1,20,000)
Contribution	60,000	60,000

for case a) Profit = $90,000 - 80,000 = 10,000$

b) Profit = $1,20,000 - 80,000 = 40,000$

16. Calculation of Absorption Costing

Absorption rate = $\frac{\text{Variable production cost per unit} + \text{Fixed production cost per unit}}{\text{Units}}$

= $4 + \frac{1500}{1500} = 25/\text{unit}$

38
x 4
22

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Particulars	Period 1	Period 2
Sales [1.200 X 6] [2.100 X 6]	7200	10,800
less: Cost of Goods Sold		
Opening Stock	-	1500
(+) Production Cost [1.1500 X 5]	7500	7500
(-) Closing Stock [1.300 X 5]	(1500)	-
COGS	6000	9000
Gross Profit	1200	1800
less: Other fixed cost	(500)	(500)
Profit	700	1300

Calculation under Marginal Costing

Particulars	Period 1	Period 2
Sales	7200	10,800
(-) V.C [4]	(4800)	(7200)
Contribution	2400	3600
(-) FC	(2000)	(2000)
EBIT	400	1600

	Period 1	Period 2
Profit as per Marginal Costing	400	1600
Add: Over-absorption of Fixed oh (carried in closing stock)	300	
less: Under-absorption of Fixed oh (released from op stock)		(-300)
Profit as per Absorption Costing	700	1300

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18. Fixed Production/unit:

$$\frac{16,00,000}{12,80,000} = ₹ 1.25/\text{unit}$$

$$\therefore \text{Absorption rate} = \text{Variable prodn o/h} + \text{Fixed prodn o/h}$$

$$\therefore (40 + 16 + 10) + 1.25 = ₹ 67.25/\text{unit}$$

* All info - Quarterly Basis

I. Marginal Costing

Statement of P&L using MC [Jan-Mar]

Particulars	(₹)
Sales (2,40,000 X 180)	4,32,00,000
(-) VC (2,40,000 X 66)	(1,58,40,000)
Contribution	2,73,60,000
(-) FC	
Fixed prodn o/h	(4,00,000)
[16,00,000 X 1/4]	(3,20,000)
Fixed selling exp	
Profit	2,66,40,000

II. Absorption Costing

Sales (2,40,000 X 180)	4,32,00,000
(-) COGS	

Opening Stock	
(+) Prodn Cost	1,88,30,000
[67.25 x 2,80,000]	
(-) Closing Stock	(26,90,000)
[67.25 x 40,000]	
G/P	(1,61,40,000)
(-) Fixed costs (selling)	2,70,60,000
	(3,20,000)
N/P	267,40,000
less: Unabsorbed fixed o/h	(50,000)
	2,66,90,000

Note: M.C expenses: the entire 4Lakh whereas A.C expenses 3.5L in current period & Carries forward ₹50,000 in inventory. That is why absorption costing is still 50,000 higher despite deducting ₹50,000 as under absorbed o/h

Reconciliation Statement

Profit as per M.C	2,66,40,000
Add: Fixed prodn o/h incl. in closing stock	50,000
Profit as per A.C	2,66,90,000

20. Calculation of Closing Stock for 3 months (in units)

Particulars	Month 1	Month 2	Month 3
Opening Stock		1500	8000
(+) Prodn	6500	9000	10,000
(-) Sales	(5000)	(8500)	(9500)
Closing Stock	1500	9000	2500

Income Statement under Marginal costing

Particulars	MONTH 1	MONTH 2	MONTH 3
Sales (₹30/unit)	1,50,000	2,55,000	2,85,000
(-) VC			
(+) Opening stock	-	34,500	46,000
(+) Production cost (10+5+8)	1,49,500	2,09,000	23,000
(-) Closing Stock (1500 x 23)	(34,500)	(46,000)	(57,500)
COGS	1,15,000	1,95,500	2,18,500
Contribution	35,000	59,500	22,500
(-) Fixed Cost	(27,000)	(22,000)	(27,000)
Profit	8000	32,500	39,500

Income Statement under Absorption Costing

Particulars	Month 1	Month 2	Month 3
Sales (₹30/unit)	1,50,000	2,55,000	2,85,000
(-) VC COGS			
Op stock	-	39,000	52,000
(+) Production Cost (variable)	(1,69,000)	2,34,000	2,60,000
C23 + (22,000/1000) 3 = 26			
(-) Closing Stock	(39,000)	(52,000)	(65,000)
Profit	20,000	34,000	38,000
Under/over absorption of FC	(7500)		3000
N/P	12,500	34,000	41,000

Reconciliation Statement

Particulars	Month 1	Month 2	Month 3
Profit as per M.C	8000	32,500	39,500
Add: Fixed Manufacturing Cost inc. Clo. Stock	4500	4500	1500
Less: Fixed Man. Cost inc in Q Stock		(4500)	(6000)
Profit as per A.C	12,500	34,000	41,000

3. CVP Analysis

1. Calculation of Revised Variable Cost / unit

Particulars	Amt (₹)
Direct Material	4.20
Direct Wages	1.05
Factory O/h	3.00
Selling O/h	0.25
Total Variable Cost	8.50
Revised Fixed Cost	2,47,500

Statement of Profit (Evaluation of new offer)

Particulars	Existing Market	New Offer	Total
(-) Selling Price	14.50	x	
(-) Variable Cost	8.50	8.50	17.00
Contribution	6.00	x - 8.50	
Total Contribution	3,60,000	20,000x - 1,70,000	
(-) Fixed Cost	(2,47,500)	(2,47,500) -	
Profit	1,12,500	20,000x - 1,70,000	1,60,000

$$1,12,500 + 20,000x - 1,70,000 = 1,60,000$$

$$+ 2,17,500 = + 20,000x$$

$$x = 10.875$$

∴ Min selling price / unit recommended for sale of additional 20,000 units to make an overall profit of ₹1,60,000 is ₹10.875

2. Particulars

	Present	I Present Export Sales	II Export Sales Chain Sales	SP ↓ 5	SP ↓ 8	SP ↓ 11
Selling price	75	60	55	70	67	64
(-) Variable Cost p.u	(50)	(55)	(50)	(50)	(50)	(50)
Contribution p.u	25	5	5	20	17	14
No of units	40,000	30,000	30,000	50,000	70,000	75,000
Total Contribution	10,00,000	1,50,000	1,50,000	10,00,000	11,90,000	10,50,000
(-) Fixed Cost	10,00,000	—	—	(10,00,000)	(10,00,000)	(10,00,000)
	—	1,50,000	1,50,000	—	1,90,000	50,000

Proposal 3 with the 2nd alternative is a better option to improve the profitability by reducing the SP/unit to 67 & increasing the sales level to 70,000 units

3. Calculation of Cost of purchase & cost of production:

Particulars	Buy	Make
Cost of purchase	4.85	6.25
Materials	-	2.75
Labour	-	1.75
Other variables	1.25 -	0.5
Fixed Costs	<u>1.25</u>	<u>1.25</u>
Total	6.10	6.25

a) $\therefore$ Buy product

b) $\therefore$ Cost = $5.85 + 1.25 = 7.10$

$\therefore$ Make product

4. Calculation of Profit

Particulars	2000 U	3000 U
Material Sales	4000 1,60,000	1,95,000
Variable (Mat + wages)	(76,000)	(76,000) (1,14,000)
Factory o/h	(20,000)	(20,000)
Selling & Distr. Packaging Cost	(16,000)	(24,000)
Contribution	48,000	24,000 24,000 (+3000)
(-) Fixed o/h	(40,000)	-
Profit	8000	24,000

$\therefore$ Accept foreign order.

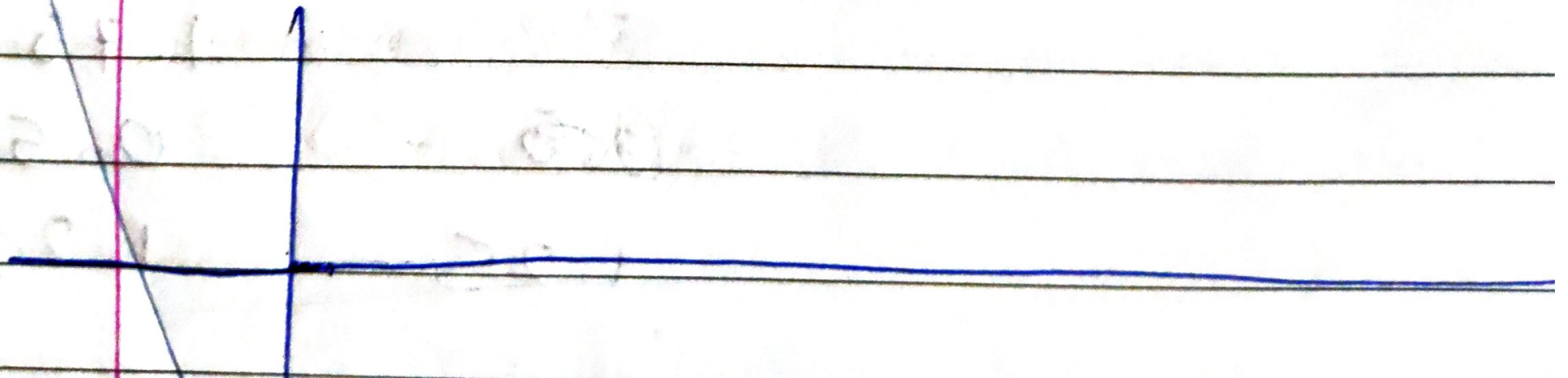
5. Sales at various capacities

@ 80% 32L

@ 100% 40L

@ 50% 20L

@ 50% (Overseas 10% & SP) 18L



5.

Particulars	Alt I	Alt II		
	D. Market 80%	D. Market 50%	O. Small 90%	Total
Total Sales	32,00,000	20,00,000	18,00,000	38,00,000
(-) V.C.				
Materials	(10,00,000)	(6,25,000)	(6,25,000)	(12,50,000)
Labour	(4,00,000)	(2,50,000)	(2,50,000)	(5,00,000)
Variable o/h	(2,00,000)	(1,25,000)	(1,25,000)	(2,50,000)
Contribution	16,00,000	10,00,000	8,00,000	18,00,000
(-) FC	(13,00,000)			(13,00,000)
Profit	3,00,000			5,00,000

Alternate III

Particulars	D. Market 80%	O. S. Market 30%	O. S. Market 20%	Total
Sales	32,00,000	10,80,000	7,20,000	50,00,000
(-) V.C.				
Materials	10,00,000	3,75,000	2,50,000	
Labour	4,00,000	1,50,000	1,50,000	
Variable o/h	2,00,000	75,000	50,000	
Contribution	16,00,000	4,80,000	3,70,000	24,50,000
(-) FC				(14,00,000)
Profit				9,50,000

6	Particulars	Present	Option 1	Option 2
	Sales	50,00,000	71,25,000	49,50,000
	(-) V.C			
	Raw mat	(20,00,000)	(30,00,000)	(18,00,000)
	Labour	(6,00,000)	(9,00,000)	(5,40,000)
	Variable costs	(4,00,000)	(6,00,000)	(3,60,000)
	Total Variable Cost	30,00,000	45,00,000	27,00,000

Total Contribution	20,00,000	26,25,000	22,50,000
Less: Fixed Cost			
Non Costs	(7,00,000)	(7,00,000)	(6,50,000)
Gen o/h	(3,00,000)	(3,00,000)	(2,00,000)
Add costs		(1,50,000)	
Profit	10,00,000	14,75,000	14,00,000

Decision: Choose Option 2

7-	Particulars	A	B
	Selling Price	20	15
	(-) V.C		
	Materials	(10)	(9)
	Direct wages	(3)	(2)
	Variable exp	(3)	(2)
	Total	14	2

Particulars	Sales Mix 1		Total
	A	B	
Contribution p.u	4	2	
No of units	1000	2000	
Total Contrib	4000	4000	8000
(-) FC			800
Profit			7200

Particulars	Sales mix 2		Total
	A	B	
Contri p.u	4	2	
No of units	1500	1500	
Total Contri	6000	3000	9000
(-) FC			(800)
Profit			8200

Particulars	Sales Mix 3		Total
	A	B	
Contri p.u	4	2	
No of units	2000	1000	
Total Contri	8000	2000	10,000
(-) FC			(800)
Profit			9200

Go with sales mix 3 because it yields higher profits.

8	Particulars	A	B	C	Total
	Sp	140	120	90	350
	(-) VC	(45)	(55)	(50)	(230)
	Contri	45	35	40	120
	X units	6400	3200	2900	
	Contri	2,88,000	1,12,000	96,000	196,000
	(-) FC	(57,600)	(70,400)	(43,200)	
		2,30,400	4,16,000	52,800	

Total Profit = 324,800

A	B	C
80:20	40:20	20:20
4kg	2kg	1kg

	A	B	C
Contri/unit	45	35	40
kg/unit	4	2	1
	11.25	17.5	40
	③	②	①

Raw material is a limiting factor

Prioritize C, then B, then A

Calculation of optimal product mix

Product mix	kg/unit	no of unit	Total kgs
Available			19,400
To C:	1	2400	18,400 (2400)
To B:	2	3200	(12,800) / 6400
To A	4	2400	9600

Particulars	A	B	C
Contri/unit	45	35	40
X units	X 2400	X 3200	X 2400
Total Contri	1,08,000	1,12,000	96,000
(-) FC	(52,600)	(70,400)	(43,200)
	50,400	41,600	52,800

10. Key Factor or Limiting Factor

Statement of Profit			
Particulars	Product A	Product B	Total
S.P p.u	14	11	
Variable Cost p.u			
Materials	5	5	
Labour	3	1.5	
Variable o/h	1.5	1.5	
Total Variable Cost p.u	9.5	8	
Contribution p.u	4.5	3	
No of units	500	600	
Total Contribution	2250	1800	4050
(-) Fixed Cost	750	450	1200
Profit	1500	1350	2850

Ranking of Products

Particulars	Product A	Product B
Contribution p.u	4.50	3
Labour hrs / product	6	3
Contri / labour hr	0.75	1.00
Ranking	2nd	1st

Optimal product mix

Particulars	No of units	Labour hrs	Total
Product A	400	6	2400
Product B	600	3	1800
			4200

9. Particulars	A	B
SP	75	48
(-) VC	(45)	(32)
Contri / unit	30	16
÷ Labour hrs	÷ 15	÷ 2
Contri / Lab hrs	2	8
Rank	II	I

During labour shortage, the firm should prioritize supplying labour to the product making max contri / lab hrs.

10th question continued.....

Particulars	A	B	Total
S.P	14	11	
Variable cost p.u			
Materials	5	5	
Labour	3	1.5	
Variable o/h	1.5	1.5	
Contri p.u	4.5	3.0	
X no of units	400	600	
Total contri	1800	1800	
(-) FC	(750)	(450)	
Profit	1050	1350	

11. Particulars	(80K)	(20K)	Local
	Local Merch	Foreign	Total
Sales	40,00,000	7,20,000	
(-) VC (12+9+6)	(21,60,000)	(5,40,000)	
Contribution	19,20,000	1,80,000	
Contri / unit (-) FC	(14,40,000)	-	
Profit	4,80,000	1,80,000	6,60,000

∴ Order can be accepted since profit is increased with increase in production within the capacity

Reject order from local merchant if it is affecting the normal local price

12. Particulars	Continue	Shutdown
Sales	70,000	-
(-) V.C	5000x	-
Contribution	70,000 - 5000x	-
(-) FC	(30,000)	(12,000)
Profit	40,000 - 5000x	(12,000)

$$40,000 - 5000x = -12,000$$

$$52,000 = 5000x$$

$$x = 10.4$$

If output > shutdown point → continue business ops
 output = shutdown point → Indifference point
 output < shutdown, close ops

$$\star \text{ Shutdown point} = \frac{\text{Available fixed cost}}{\text{Contribution/unit}}$$

13. $SP = 8$
 $VC = (6)$

Contri 2
 units 95,000 units
 Contri 1,90,000

FC for running ops = 3,50,000
 FC if shutdown = 1,30,000 + 15,000 = 1,45,000
 Avoidable FC = 3,50,000 - 1,45,000
 = 2,05,000

Shutdown point = $\frac{2,05,000}{2} = 1,02,500$ units

Present level of output = 95,000

The present output < shutdown point

(Immediate shut down), since present output is less than the SDP, it is suggested to shut down the plant

14. Present output = 2,00,000 units (normal capacity)
 next quarter = 10,000 units

$SP = 21$
 $VC = (7.8 + 2.1 + 2.5 + 0.60) = 13$
 Contri 8

Fixed o/h if continued = $\frac{4 \times 2,00,000}{4} = 2,00,000$ for one quarter

FC if shutdown = 74,000 + 14,000 = 88,000

Avoidable FC = 2,00,000 - 80,000
 = 1,20,000

Shutdown point = $\frac{1,20,000}{8} = 14,000$ units

$14,000 > 10,000$ units $\Rightarrow$ more losses

Since activity level is only for 10,000 units, i.e. below SDP of 14,000 units, the plant may be closed for the quarter.

Above 14,000 units, ops may continue

Losses if continued would be 1,20,000 $[(8 \times 10,000) - 21]$

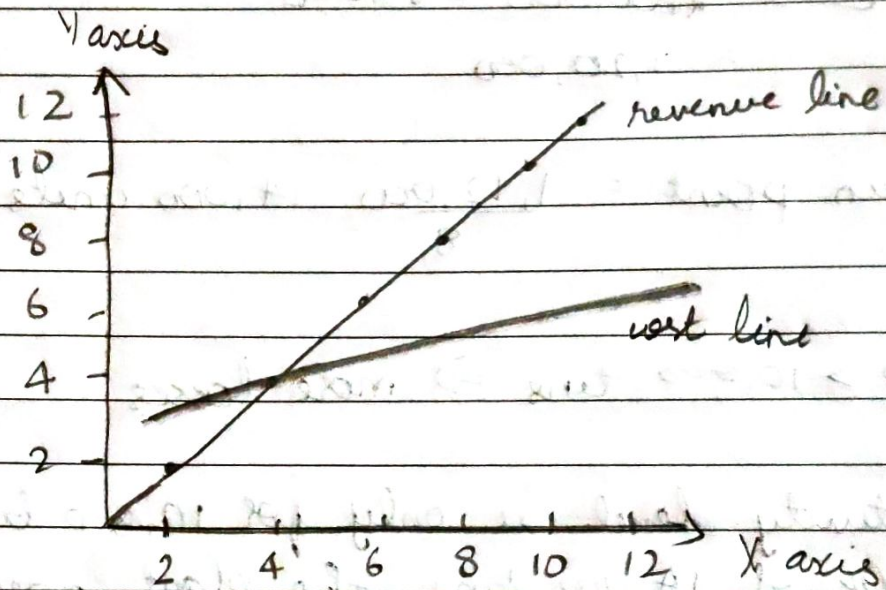
Loss if shut down would be 88,000
 $[74,000 + 14,000]$

Break-even by PV chart

15. Basic Computation

Particulars	2000 u	4000 u	6000 u	8000 u	10,000 u
SP/unit (₹)	2000	4000	6000	8000	10,000
(-) VC (0.5)	(1000)	(2000)	(3000)	(4000)	(5000)
Contribution	1000	2000	3000	4000	5000
(-) FC	(2000)	(2000)	(2000)	(2000)	(2000)
	(1000)	-	1000	2000	3000

$$BEP = \frac{FC}{\text{Contribution / unit}} = \frac{2000}{0.5} = 4000 \text{ units}$$



16. Basic Computation

Particulars	50K	1L	1.5L	2L	2.5L
SP (20)	10L	20L	30L	40L	50L
VC (10)	(5L)	(10L)	(15L)	(20L)	(25L)
Contri	5L	10L	15L	20L	25L
FC (5L)	(5L)	(5L)	(5L)	(5L)	(5L)
	0 -	- 5L	10L	15L	20L